

1. Bayesian Regime Detection Engine: Problem & Objective

Institutional Platform Architecture, Probabilistic Calibration & Audit Defense

- Organization: Zetheta Algorithms Private Limited (CIN: U62012MH2023PTC410415).
- Target Canonical Environment: Python 3.10 | Host Platform: macOS ARM64 / Linux x86_64.
- The Problem: Financial price series have signal-to-noise ratio < 0.05 ; point forecasting fails out-of-sample.
- The Objective: Predict probability distributions over 5 macroeconomic regimes for tactical equity allocation.

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2. The Direction-Over-Price Thesis

Why Point Return Predictions Fail in Quantitative Asset Allocation

- Martingale Diffusion: $E[R_{t+1}|F_t] = \mu_t \ll \sigma_t$; expected return is dwarfed by daily volatility.
- Asymmetric Fat Tails: NIFTY 50 exhibits kurtosis of 19.04 and skewness of -0.68, refuting Gaussian assumptions.
- False Precision: Point forecasts assert misleading certainty, causing aggressive turnover and friction losses.
- Paradigm Shift: Predict conditional regime probabilities $P(S_t = k | X_t)$ on the unit simplex Δ^5 .

3. The Canonical Five-Regime Framework

Empirical Macro Dynamics Across Indian Equities & Target Definition

- Canonical Ontology: Risk-On (Bull Quiet), Late-Cycle (Bull Volatile), Transitional, Post-Shock, Risk-Off (Bear Panic).
- Evaluation Proxy: 5-day forward NIFTY return and realized volatility against training median threshold.
- Non-Circular Rule: Forward target is an objective outcome proxy, not an unobserved daily latent ground truth.
- RPS Monotonic Ordering: [Risk-On] < [Late-Cycle] < [Post-Shock] < [Transitional] < [Risk-Off] based on return/risk severity.

4. Data Universe & Point-in-Time Architecture

15 Years of Indian Equity History (2009–2024) with Zero Lookahead

- Data Scale: 35,766 OHLCV bars across 3,949 trading days covering NIFTY 50, Midcap 50, Bank, IT, VIX, USD/INR.
- Point-in-Time Guards: Strict lag enforcement ensuring feature calculation uses strictly backward information.
- Cryptographic Hashes: Market data SHA-256 (c6c46ed7...) and Feature store SHA-256 (61969b7b...).
- Quarantine Policy: Macro regulatory portals (RBI, SEBI, AMFI) quarantined under zero synthetic data rule.

5. Feature Store: Technical, Topological & Graph Features

30 Point-in-Time Engineered Features Across Three Paradigms

- Technical (16): Momentum returns (1d, 5d, 21d), EWMA volatility ($\lambda=0.94$), Parkinson vol, moving average trend distances.
- Topological Data Analysis (5): Vietoris-Rips persistent homology (H_0/H_1 persistence entropy, Wasserstein amplitude).
- Dynamic Graph Neural Network (9): Rolling sector correlation Laplacian, Fiedler algebraic connectivity (λ_2).
- Stationarity Audit: 28 of 30 features stationary at $p < 0.001$ (ADF test); Multicollinearity controlled via $VIF < 10$.

6. Sticky Dirichlet Bayesian HMM & PyMC NUTS Sampler

Preventing Artificial Regime Whipsaws via Informative Priors

- Sticky Prior: Self-transition concentration $A_{\{j,j\}} \sim \text{Dir}(\alpha_0 + \kappa)$ with $\kappa = 8.0$.
- FFBS MCMC: 2 chains, 120 iterations, 40 burn-in via Forward-Filtering Backward-Sampling ($R\text{-hat} < 1.05$, $\text{ESS} > 70$).
- PyMC NUTS Specification Run: 4 chains, 2,000 draws, 1,000 tune, $\text{target_accept}=0.90$ executed locally.
- NUTS Diagnostics: Max $R\text{-hat} = 1.0023$, Bulk ESS = 2,875.4, Tail ESS = 2,476.8, Divergent transitions = 0 (0.00%).

7. Markov-Switching VAR & Dynamic Cross-Asset Feedback

Capturing Endogenous Regime Transmission & Duration Dynamics

- RS-VAR(1) Model: $Y_t = \nu(S_t) + \Phi(S_t) * Y_{t-1} + e_t$ for returns, VIX changes, and breadth momentum.
- Filtering & Smoothing: Hamilton (1989) forward filter and Kim (1994) full-sample backward smoother (Log-Lik: 44,816.43).
- Weibull Duration Hazard: Risk-Off duration exhibits negative duration dependence (Weibull $\alpha = 0.68$).
- Memoryless Null Rejected: Geometric dwell-time hypothesis rejected for crisis states ($p = 0.0098$).

8. Bayesian Deep Learning & Epistemic Uncertainty

Disentangling Model Ignorance from Inherent Market Noise

- Variational BNN: Bayes by Backprop in PyTorch with Gaussian weights $w \sim N(\mu, \text{softplus}(\rho))$, ELBO loss.
- Deep Ensemble: $M=3$ independently initialized neural networks with mini-batch shuffling strictly within training window.
- Monte Carlo Dropout: Preserves dropout ($p=0.20$) across 50 stochastic forward passes at inference time.
- Uncertainty Budget: Epistemic entropy (mutual information) isolates model ignorance to trigger defensive scaling.

9. Foundation Models in Macro Regime Detection: Empirical Limits

Zero-Shot Probing of Amazon Chronos T5 & Google TimesFM

- Tested Models: Amazon Chronos T5-small (context L=64) and Google TimesFM-1.0-200m (patch length 16).
- Representation Probing: Linear probing classifiers trained on causal temporal embeddings to predict 5 regimes.
- Empirical Finding: Probing accuracy is 23.77% (near 20% random baseline) with negative silhouette score (-0.0089).
- Key Insight: Generic foundation models pretrained on non-financial series fail to capture macro financial covariance.

10. Simplex Stacking Ensemble & Benchmark Tournament

Out-of-Sample Holdout (2022–2024, 738 Days) Proper-Score Audit

- Simplex Stacking: SLSQP optimization on Δ^M assigns 90.9% to Deep Ensemble and 9.1% to Bayesian HMM (Gibbs).
- Log Loss Champion: Deep Ensemble (1.2847) beats Climatology (1.3097) and Persistence (1.9083, +32.68% skill).
- Variational BNN: Achieves out-of-sample Log Loss of 1.3008 and RPS of 0.2033 (+31.84% skill vs Persistence).
- RPS Dichotomy: Persistence achieves low RPS (0.1577) due to adjacent CDF mass; all models have negative RPS skill.

11. Temperature Calibration & Adaptive Conformal Inference (ACI)

Finite-Sample Distribution-Free Coverage Guarantees Under Shift

- Temperature Scaling: Optimal $T = 1.0839$ collapses Expected Calibration Error (ECE) from 0.0350 to 0.0014 (96.1% drop).
- Conformal Guarantee: $P(Y_{t+1} \in C_{t+1}) \geq 1 - \alpha$ without parametric distributional assumptions.
- Audited Empirical Coverage: 91.33% realized holdout coverage (target 90.0%, gap +1.33%) with mean set size 3.06.
- Prudent Ambiguity: Single-regime calls drop to 0% during macro transitions, spanning multi-regime risk sets.

12. Two-Speed Real-Time Production Architecture

High-Throughput Serving via FastAPI & Bootstrap Particle Filter

- Slow Engine: Nightly batch MCMC, PyMC sampling, temperature re-fitting, and drift monitoring.
- Fast Engine: 1,000-particle Bootstrap Particle Filter (BPF) processing intraday market updates in < 2 ms.
- Bayesian Online Changepoint Detection (BOCPD): Run-length hazard tracking ($\lambda=100.0$) for rapid jump alerts.
- Reconciliation Safety Gate: $D_{KL}(P_{\text{online}} || P_{\text{batch}}) > 0.25$ triggers automatic model risk alarm and set expansion.

13. Explainability (Permutation SHAP) & Cryptographic Lineage

Human-Interpretable Intelligence Aligned with SR 11-7

- Permutation SHAP: Real-time marginal feature attributions identifying top regime drivers (volatility, breadth, TDA).
- Deterministic Narrative: Natural language brief synthesis without LLM hallucination for Investment Committees.
- SR 11-7 Lifecycle: Candidate -> Validation -> Challenger -> Promoted Champion -> Retired state machine.
- Audit Replay: Immutable cryptographic replay linking every decision to data and feature store SHA-256 hashes.

14. Risk Modeling: Monte Carlo & Conviction-Aware Allocation

Mapping Posterior Regimes to Dynamic Equity Exposure

- Student-t Monte Carlo: 10,000-path 21-day forward simulation conditioned on posterior regime probabilities.
- Risk Budgeting: 99% 21-day Value at Risk (VaR) is -7.8%; 99% Conditional VaR (CVaR) is -9.8%.
- Conviction Overlay: Dynamic equity allocation bounded in [20%, 100%] scaled by confidence $1 - H(p)/\ln(5)$.
- Turnover Hysteresis: 4.0% no-trade deadband and 10.0% daily turnover cap prevent whipsaw execution drag.

15. Historical Case Studies & Crisis Attributions

Preserving Institutional Capital During Indian Liquidity Shocks

- 2013 Taper Tantrum: USD/INR 3-sigma spike triggered early de-risking, avoiding violent currency-driven drawdown.
- 2018 IL&FS; Shock: TDA H1 persistence entropy spiked and sector connectivity collapsed 14 days before equity breakdown.
- 2020 COVID Crash: Strategy drawdown limited to -23.82% vs Benchmark -38.44% (+14.62% capital saved, +12.96% alpha).
- 2024 Election Shock: MC Dropout flagged 65.2% epistemic entropy; ACI expanded set to 4 regimes, preventing bottom liquidation.

16. Walk-Forward Backtest & Honest Failure Analysis

Strict 2019–2024 Execution Net of 15 bps Friction (1,480 Days)

- Net Performance: CAGR 12.67%, Volatility 12.83% (vs NIFTY 50 18.32%, 29.9% vol reduction), Sharpe 0.49 vs 0.44.
- Overfitting Controls: Exactly 15 parameter configurations tried; DSR = 0.112; Single-asset PBO = 0.34 (marked PARTIAL).
- Conviction Tiers: High conviction (>70%) CAGR 14.82% (Sharpe 0.74) vs Low conviction (<50%) CAGR 6.14% (Sharpe -0.02).
- Honest Failure Modes: Discontinuous overnight gap openings and protracted range-bound trendless consolidation.

17. The RegimeLab Product Platform & Regime Arena

Production Interface & Gamified Quantitative Simulation

- Streamlit Dashboard (8 Pages): Live Monitor, SHAP Explainer, Model Lab, Replay, Risk & Allocation, Governance.
- Regime Arena: Interactive trading simulator allowing analysts to allocate capital under historical point-in-time constraints.
- Human vs Engine: Real-time benchmarking comparing human emotional trading against Bayesian conviction overlay.
- API Service: Production FastAPI service with verified endpoints (GET /regime/health, POST /regime/score).

18. Conclusion, Forensic Gaps Attestation & Roadmap

94.44% Audited Completion & Strategic Institutional Deployment

- Audited Completion: 34 of 36 requirements empirically verified (94.44%); 72/72 unit and statistical tests passing.
- Quarantined Blockers: Regulatory macro feeds (BLOCKED BY DATA) and dual-language R host runtime (BLOCKED BY ENVIRONMENT).
- R Codebase Ready: Complete depmixS4, MSwM, rstanarm/Stan, changepoint, and conformal scripts with renv.lock & Docker.
- Sign-Off: Zetheta Algorithms Private Limited | CIN: U62012MH2023PTC410415 | Production Reference Delivery.

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